Supplementary Material – Unified Online Adaptation Framework for Correlation Analysis-based Spatial Filtering Methods in SSVEP-based BCIs

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S.I. Detailed derivation of spatial filter update

In order to derive the least-squares (LS) unified framework from the calibration-based format to the online adaptation format, the key parameters in the process of solving the generalized LS problem in the LS unified framework should be first identified. Then, the spatial filters can be continuously adjusted by using these parameters.

Following the LS unified framework, the spatial filters can be solved by the alternated least squares in each trial to achieve their online adaptation. Specifically, in the i-th iteration of the k-th trial, by fixing $\mathbf{M}^{[k,i-1]}$, $\mathbf{W}^{[k,i]}$ can be obtained by

$$\arg\min_{\mathbf{W}^{[k,i]}} \|\mathbf{E}^{[k]}\mathbf{W}^{[k,i]} - \mathbf{E}^{[k]}\left(\mathbf{R}^{[k]}\right)^{-1/2}\mathbf{M}^{[k,i-1]}\|_F^2, \tag{S1}$$

where

$$\mathbf{R}^{[k]} = \left(\mathbf{K}^{[k]}\right)^T \mathbf{K}^{[k]}. \tag{S2}$$

In addition, $\left(\mathbf{R}^{[k]}\right)^{-1/2}$ can be computed by the eigen decomposition of $\mathbf{R}^{[k]}$, i.e.,

$$\left(\mathbf{R}^{[k]}\right)^{-1/2} = \mathbf{V}_{\mathbf{R}}^{[k]}\left(\mathbf{D}_{\mathbf{R}}^{[k]}\right)^{-1/2}\left(\mathbf{V}_{\mathbf{R}}^{[k]}\right)^T, \tag{S3}$$

where $\mathbf{V}_{\mathbf{R}}^{[k]}$ and $\mathbf{D}_{\mathbf{R}}^{[k]}$ are eigenvectors and eigenvalues of $\mathbf{R}^{[k]}$. By letting the gradient be zero, the optimal $\mathbf{W}^{[k]}$ can be found in the closed form as

$$\mathbf{W}^{[k,i]} = \left(\mathbf{F}^{[k]}\right)^{-1}\mathbf{F}^{[k]}\left(\mathbf{R}^{[k]}\right)^{-1/2}\mathbf{M}^{[k,i-1]}, \tag{S4}$$

where

$$\mathbf{F}^{[k]} = \left(\mathbf{E}^{[k]}\right)^T \mathbf{E}^{[k]}. \tag{S5}$$

Then, by substituting $\mathbf{W}^{[k,i]}$ into the LS unified framework, $\mathbf{M}^{[k,i]}$ can be solved by

$$\max_{\mathbf{M}^{[k,i]}} \ \mathrm{trace}\left\{\mathbf{U}_{\mathbf{P}}^{[k,i]}\mathbf{D}_{\mathbf{P}}^{[k,i]}\left(\mathbf{V}_{\mathbf{P}}^{[k,i]}\right)^T\left(\mathbf{M}^{[k,i]}\right)^T\right\}, \tag{S6}$$

where $\mathbf{U}_{\mathbf{P}}^{[k,i]}$, $\mathbf{D}_{\mathbf{P}}^{[k,i]}$, and $\mathbf{V}_{\mathbf{P}}^{[k,i]}$ are the orthogonal matrices and the diagonal matrix obtained from the singular value decomposition (SVD) of

$$\mathbf{P}^{[k,i]} = \left(\mathbf{R}^{[k]}\right)^{-1/2}\mathbf{F}^{[k]}\mathbf{W}^{[k,i]}. \tag{S7}$$

According to the Cauchy-Schwarz inequality, we can get

$$\mathbf{M}^{[k,i]} = \mathbf{U}_{\mathbf{P}}^{[k,i]}\left(\mathbf{V}_{\mathbf{P}}^{[k,i]}\right)^T. \tag{S8}$$

For the initial iteration, we can use the spatial filters obtained from the last trial to initialize $\mathbf{M}^{[k,1]}$, i.e.,

$$\mathbf{M}^{[k,1]} = \mathbf{U}_{\mathbf{P}}^{[k-1,I_{k-1}]}\left(\mathbf{V}_{\mathbf{P}}^{[k-1,I_{k-1}]}\right)^T \quad (k>1), \tag{S9}$$

where I_{k-1} denotes the total iteration number in the $(k-1)$-th trial. For the 1st trial, although $\mathbf{M}^{[1,1]}$ cannot be directly estimated by (S9), it still needs to satisfy the LS unified framework. Let us reformulate the objective function of the LS unified framework as $\|\mathbf{E}^{[1]}\left(\mathbf{R}^{[1]}\right)^{-1/2}\mathbf{M}^{[1,1]}\left(\mathbf{M}^{[1,1]}\right)^T - \mathbf{E}^{[1]}\left(\mathbf{R}^{[1]}\right)^{-1/2}\|_F^2$, it is easy to show that $\mathbf{M}^{[1,1]}$ can be computed as the orthogonal matrix obtained from the SVD of $\mathbf{E}^{[1]}\left(\mathbf{R}^{[1]}\right)^{-1/2}$. In other words, $\mathbf{M}^{[k,1]}$ in the 1st trial can be estimated by

$$\mathbf{M}^{[k,1]} = \mathbf{V}_{\mathbf{T}}^{[k]} \quad (k=1), \tag{S10}$$

where $\mathbf{V}_{\mathbf{T}}^{[k]}$ is the eigenvector of $\left(\mathbf{R}^{[k]}\right)^{-1/2}\mathbf{F}^{[k]}\left(\mathbf{R}^{[k]}\right)^{-1/2}$.

A. Online msCCA

As mentioned in [*S1*], the combined EEG features of inter-classes $\mathbf{E}$ of the msCCA can be computed as

$$\mathbf{E}_{\text{msCCA}} = \sum_{s=1}^{S}\sum_{t=1}^{T_s}\mathbf{Q}_s^T\mathbf{X}_{s,t}, \tag{S11}$$

Therefore, in the online adaptation process, suppose the new incoming data of the k-th trial belongs to the s-th stimulus, $\mathbf{E}^{[k]}$ can be updated by

$$\mathbf{E}_{\text{msCCA}}^{[k]} = \mathbf{E}_{\text{msCCA}}^{[k-1]} + \mathbf{Q}_s^T\mathbf{X}^{[k]}. \tag{S12}$$

However, in real online adaptation, the label s of the new trial data is unknown. Consequently, the pseudo-label provided by the spatial filters and templates in the $(k-1)$-th trial is used for the online adaptation, which is denoted as $\hat{s}$. Then, $\mathbf{F}^{[k]}$ can be updated by

$$\mathbf{F}_{\text{msCCA}}^{[k]} = \mathbf{F}_{\text{msCCA}}^{[k-1]} + \left(\mathbf{C}_{\mathbf{F},\text{msCCA}}^{[k]}\right)^T\mathbf{O}_{\mathbf{F},\text{msCCA}}\mathbf{C}_{\mathbf{F},\text{msCCA}}^{[k]}, \tag{S13}$$

where

$$\mathbf{C}_{\mathbf{F},\text{msCCA}}^{[k]} = \begin{bmatrix}\mathbf{E}_{\text{msCCA}}^{[k-1]}\\ \mathbf{Q}_{\hat{s}}^T\mathbf{X}^{[k]}\end{bmatrix},\ \mathbf{O}_{\mathbf{F},\text{msCCA}} = \begin{bmatrix}\mathbf{0} & \mathbf{I}\\ \mathbf{I} & \mathbf{I}\end{bmatrix}, \tag{S14}$$

and $\mathbf{I}$ is the identity matrix.

Furthermore, according to [*S1*], $\mathbf{R}$ of the msCCA is computed as

$$\mathbf{R}_{\text{msCCA}} = \sum_{s=1}^{S}\left\{\left(\sum_{t=1}^{T_s}\mathbf{X}_{s,t}\right)^T\left(\sum_{t=1}^{T_s}\mathbf{X}_{s,t}\right)\right\}. \tag{S15}$$

Therefore, in the online adaptation, $\mathbf{R}^{[k]}$ can be updated by

$$\mathbf{R}_{\text{msCCA}}^{[k]} = \mathbf{R}_{\text{msCCA}}^{[k-1]} + \left(\mathbf{C}_{\mathbf{R},\text{msCCA}}^{[k]}\right)^T\mathbf{O}_{\mathbf{R},\text{msCCA}}\mathbf{C}_{\mathbf{R},\text{msCCA}}^{[k]}, \tag{S16}$$

where

$$\mathbf{C}_{\mathbf{R},\text{msCCA}}^{[k]} = \begin{bmatrix}\mathbf{\Sigma}_{\hat{s}}^{[k-1]}\\ \mathbf{Q}_{\hat{s}}^T\mathbf{X}^{[k]}\end{bmatrix},\ \mathbf{\Sigma}_{\hat{s}}^{[k]} = \mathbf{\Sigma}_{\hat{s}}^{[k-1]} + \mathbf{X}^{[k]},\ \text{and}$$

$$\mathbf{O}_{\mathbf{R},\text{msCCA}} = \mathbf{O}_{\mathbf{F},\text{msCCA}} \tag{S17}$$

Finally, after tuning the spatial filter, the harmonic weights of the sine-cosine reference signals can be updated by

$$\mathbf{U}^{[k]} = \mathbf{E}_{\text{msCCA}}^{[k]}\mathbf{W}^{[k]}. \tag{S18}$$

It should be noticed that, suppose that $\mathbf{X}^{[k]}$ is the 1st trial that is recognized as the s-th stimulus, $\mathbf{\Sigma}_{\hat{s}}^{[k-1]}$ will be $\mathbf{0}$. In addition, $\mathbf{\Sigma}_{\hat{s}}^{[k]}$ will be updated only when the pseudo-label of the new coming trial is $\hat{s}$, and remain unchanged in other cases. Moreover, although the spatial filter obtained from the online msCCA proposed in this study is exactly the same as that obtained from the OMSCCA proposed in [*S2*], this study covers the online adaptation processes of the msCCA into the unified online adaptation framework, which would be helpful for exploring the relationships between the online adaptation of the msCCA and other advanced CA-based spatial filtering methods introduced in the following sections.

B. Online msTRCA

Comparing $\mathbf{Z}$, $\mathbf{L_E}$, $\mathbf{P_E}$, $\mathbf{L_K}$, and $\mathbf{P_K}$ of the msCCA and msTRCA shown in [*S1*], we can easily find that $\mathbf{E}$ of the msTRCA is the same as $\mathbf{K}$ of the msCCA, i.e.,

$$\mathbf{F}_{\text{msTRCA}} = \mathbf{R}_{\text{msCCA}}. \tag{S19}$$

In other words, $\mathbf{F}^{[k]}$ can be adapted online following (S16), i.e.,

$$\mathbf{F}_{\text{msTRCA}}^{[k]} = \mathbf{F}_{\text{msTRCA}}^{[k-1]} + \left(\mathbf{C}_{\mathbf{F},\text{msTRCA}}^{[k]}\right)^T\mathbf{O}_{\mathbf{F},\text{msTRCA}}\mathbf{C}_{\mathbf{F},\text{msTRCA}}^{[k]}, \tag{S20}$$

where

$$\mathbf{C}_{\mathbf{F},\text{msTRCA}}^{[k]} = \mathbf{C}_{\mathbf{R},\text{msCCA}}^{[k]}\ \text{and}\ \mathbf{O}_{\mathbf{F},\text{msTRCA}} = \mathbf{O}_{\mathbf{R},\text{msCCA}} \tag{S21}$$

Moreover, since $\mathbf{R}$ of the msTRCA in the LS unified framework is computed by

$$\mathbf{R}_{\text{msTRCA}} = \sum_{s=1}^{S}\sum_{t=1}^{T_s}\mathbf{X}_{s,t}^T\mathbf{X}_{s,t}, \tag{S22}$$

the corresponding online adaptation can be achieved by

$$\mathbf{R}_{\text{msTRCA}}^{[k]} = \mathbf{R}_{\text{msTRCA}}^{[k-1]} + \left(\mathbf{C}_{\mathbf{R},\text{msTRCA}}^{[k]}\right)^T \mathbf{O}_{\mathbf{R},\text{msTRCA}} \mathbf{C}_{\mathbf{R},\text{msTRCA}}^{[k]}, \tag{S23}$$

where

$$\mathbf{C}_{\mathbf{R},\text{msTRCA}}^{[k]} = \mathbf{X}^{[k]}, \text{ and } \mathbf{O}_{\mathbf{R},\text{msTRCA}} = \mathbf{I}. \tag{S24}$$

C. Online TDCA

For the TDCA, the newly arriving EEG signals should be first augmented across channels by including delay signals, and then extended across samples by combining the original EEG signals and the corresponding EEG features. According to the LS unified framework, $\mathbf{F}$ of the TDCA should be computed as

$$\mathbf{F}_{\text{TDCA}} = \frac{1}{S}\sum_{s=1}^{S}\left\{\left(\frac{1}{T_s}\sum_{t=1}^{T_s}\hat{\mathbf{X}}_{s,t,d} - \frac{1}{S}\sum_{s=1}^{S}\frac{1}{T_s}\sum_{t=1}^{T_s}\hat{\mathbf{X}}_{s,t,d}\right)^T \times \left(\frac{1}{T_s}\sum_{t=1}^{T_s}\hat{\mathbf{X}}_{s,t,d} - \frac{1}{S}\sum_{s=1}^{S}\frac{1}{T_s}\sum_{t=1}^{T_s}\hat{\mathbf{X}}_{s,t,d}\right)\right\}. \tag{S25}$$

We can easily find that the conventional TDCA assumes that the total trial numbers are the same for all stimuli, and uses the averaged augmented EEG signals to compute the spatial filter. However, in real online process, this assumption may not be satisfied. In addition, using the average of trials equivalently increases the weights of trials in stimuli containing less trials, and thus makes the spatial filter update process unstable. Therefore, we cannot directly generate (S25) to produce the online TDCA.

By multiplying the trial number and the stimulus number to both sides of the equation in the unified LS framework, we can further derive the computation of $\mathbf{F}$ in the TDCA as

$$\mathbf{F}_{\text{TDCA}} = \sum_{s=1}^{S}\left\{\left(S\sum_{t=1}^{T_s}\hat{\mathbf{X}}_{s,t,d} - \sum_{s=1}^{S}\sum_{t=1}^{T_s}\hat{\mathbf{X}}_{s,t,d}\right)^T \times \left(S\sum_{t=1}^{T_s}\hat{\mathbf{X}}_{s,t,d} - \sum_{s=1}^{S}\sum_{t=1}^{T_s}\hat{\mathbf{X}}_{s,t,d}\right)\right\}. \tag{S26}$$

We can find that, in this summation format, for the stimuli containing more trials, more signals are summed together in $\sum_{t=1}^{T_s}\hat{\mathbf{X}}_{s,t,d}$. Since all signals are normalized in the pre-process, the summation of more signals will give higher weights to $\sum_{t=1}^{T_s}\hat{\mathbf{X}}_{s,t,d}$. Therefore, we do not need to design specific update strategies for the trial weights. The online adaptation process generated from (S26) can automatically provide higher stimulus weights for the stimuli containing more trials. In other words, although deriving (S26) is based on the assumption that the total trial numbers are the same for all stimuli, extending (S26) to the online adaptation will equivalently abandon this assumption, and apply different trial numbers to different stimuli.

Based on (S26), $\mathbf{F}^{[k]}$ can be updated by

$$\mathbf{F}_{\text{TDCA}}^{[k]} = \mathbf{F}_{\text{TDCA}}^{[k-1]} + \left(\mathbf{C}_{\mathbf{F},\text{TDCA}}^{[k]}\right)^T \mathbf{O}_{\mathbf{F},\text{TDCA}} \mathbf{C}_{\mathbf{F},\text{TDCA}}^{[k]}, \tag{S27}$$

where

$$\mathbf{C}_{\mathbf{F},\text{TDCA}}^{[k]} = \begin{bmatrix}\hat{\mathbf{\Sigma}}^{[k]} \\ \hat{\mathbf{\Sigma}}_{\hat{s}}^{[k]} \\ \hat{\mathbf{X}}^{[k]}\end{bmatrix} \text{ and}$$

$$\mathbf{O}_{\mathbf{F},\text{TDCA}} = \begin{bmatrix}\mathbf{0} & \mathbf{0} & -S\cdot\mathbf{I} \\ \mathbf{0} & \mathbf{0} & S^2\cdot\mathbf{I} \\ -S\cdot\mathbf{I} & S^2\cdot\mathbf{I} & S(S-1)\cdot\mathbf{I}\end{bmatrix}, \tag{S28}$$

as well as $\hat{\mathbf{\Sigma}}^{[k]} = \hat{\mathbf{\Sigma}}^{[k-1]} + \hat{\mathbf{X}}^{[k]}$, and $\hat{\mathbf{\Sigma}}_{\hat{s}}^{[k]} = \hat{\mathbf{\Sigma}}_{\hat{s}}^{[k-1]} + \hat{\mathbf{X}}^{[k]}$. Similar to $\mathbf{\Sigma}_{\hat{s}}^{[k]}$ of the online msCCA, $\hat{\mathbf{\Sigma}}_{\hat{s}}^{[k]}$ will also be updated only when the new coming trial is recognized as the s-th stimulus.

Furthermore, according to the $\mathbf{Z}$, $\mathbf{L_K}$, and $\mathbf{P_K}$ of the TDCA in the LS unified framework [*S1*], $\mathbf{K}$ is constructed by vertically concatenating the detrended signals in all trials. Therefore, the online adaptation of $\mathbf{R}^{[k]}$ can be computed by

$$\mathbf{R}_{\text{TDCA}}^{[k]} = \mathbf{R}_{\text{TDCA}}^{[k-1]} + \left(\mathbf{C}_{\mathbf{R},\text{TDCA}}^{[k]}\right)^T \mathbf{O}_{\mathbf{R},\text{TDCA}} \mathbf{C}_{\mathbf{R},\text{TDCA}}^{[k]}, \tag{S29}$$

where

$$\mathbf{C}_{\mathbf{R},\text{TDCA}}^{[k]} = \begin{bmatrix}\hat{\mathbf{\Sigma}}_{\hat{s}}^{[k]} \\ \hat{\mathbf{X}}^{[k]}\end{bmatrix} \text{ and}$$

$$\mathbf{O}_{\mathbf{R},\text{TDCA}} = \begin{bmatrix}\mathbf{I} & -(T_{\hat{s}}^{[k]}-1)\cdot\mathbf{I} \\ -(T_{\hat{s}}^{[k]}-1)\cdot\mathbf{I} & (T_{\hat{s}}^{[k]}-1)^2\cdot\mathbf{I}\end{bmatrix}. \tag{S30}$$

Suppose the new coming trial is recognized as the j-th trial of the s-th stimulus, the number of total trials of the s-th stimulus is j in the current trial, which means that $T_{\hat{s}}^{[k]} = j$.

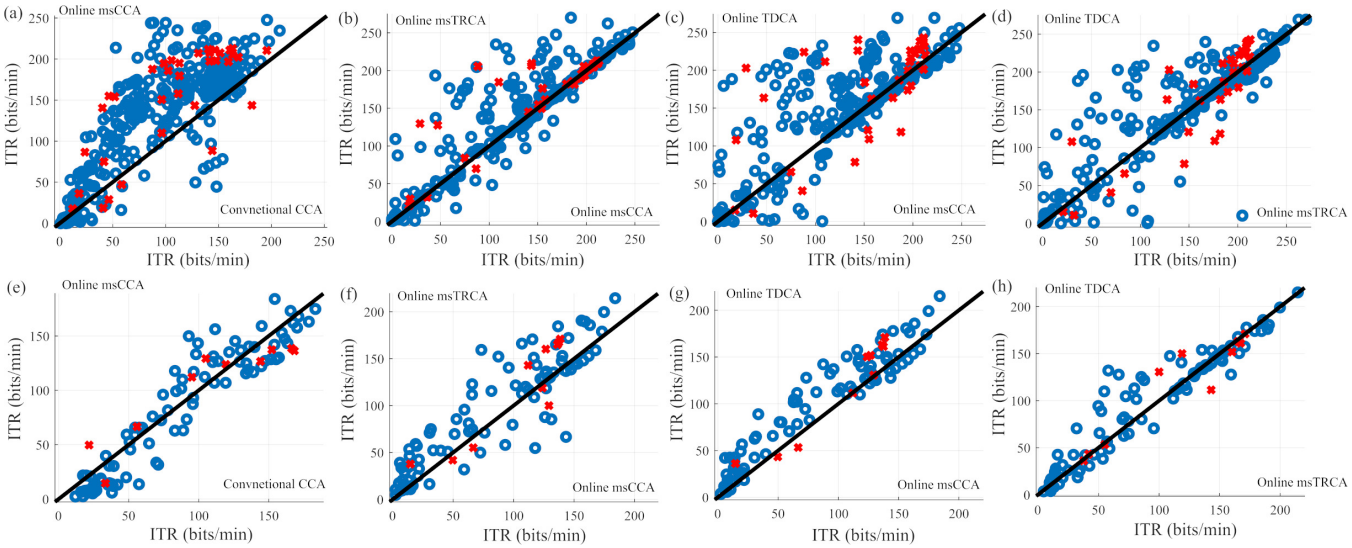

Fig. S1. Comparisons of individual ITR performance of the conventional CCA and three proposed online adaptation methods on the Benchmark Dataset and the UCSD Dataset. (a) and (e) show the comparisons between the conventional CCA and the online msCCA. (b) and (f) show the comparisons between the online msTRCA and the online msCCA. (c) and (g) show the comparisons between the online TDCA and the online msCCA. (d) and (h) show the comparisons between the online TDCA and the online msTRCA. (a)–(d) are results on the Benchmark Dataset. (e)–(h) are results on the UCSD Dataset. The red cross points denote the performance of the signal lengths that the corresponding methods achieve their highest average ITR values.

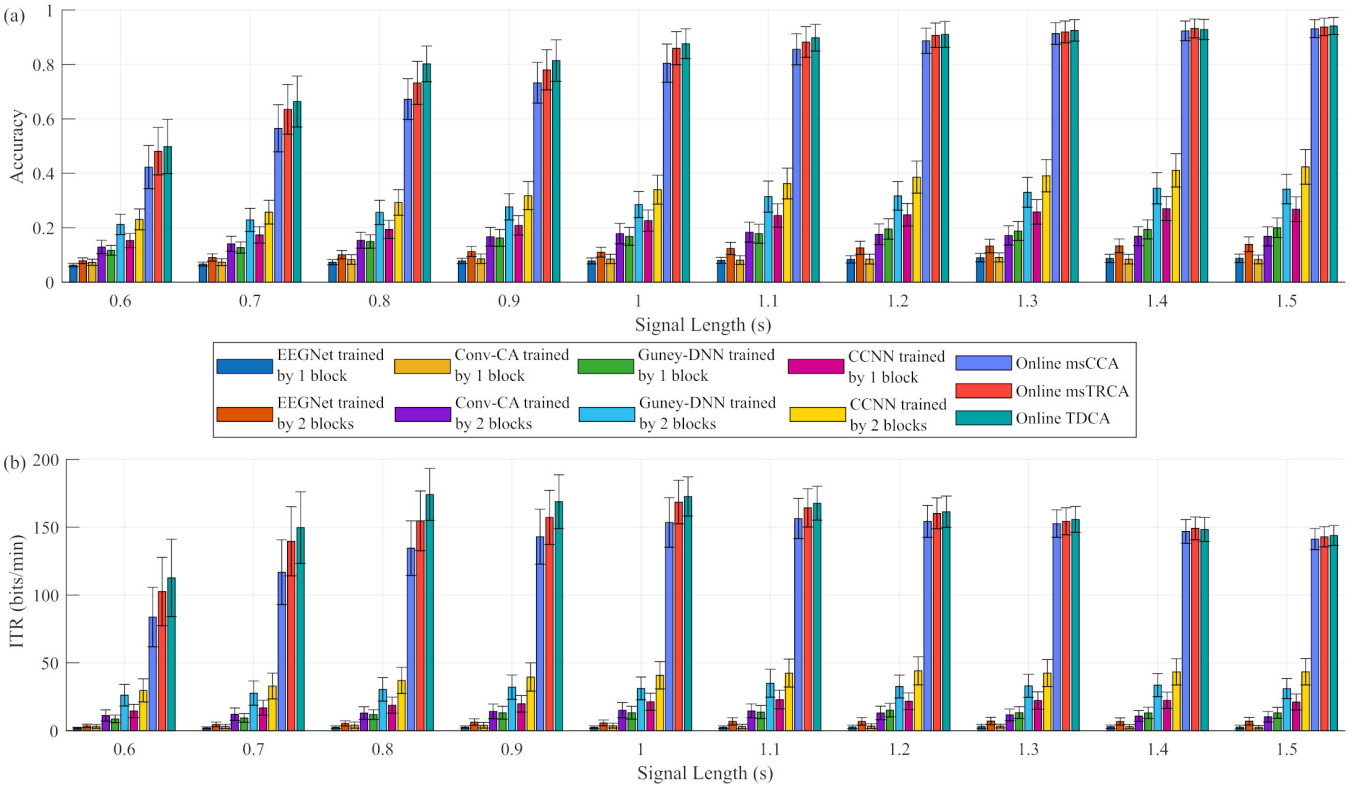

Fig. S2. Recognition performance of three proposed online adaptation methods as well as four deep learning-based methods calibrated by 1 block and 2 blocks on the Benchmark Dataset. (a) Average accuracy performance. (b) Average ITR performance.

S.II. Detailed Recognition Performance Comparisons of Whole Trials

Fig. S1 illustrates the individual ITR performance of the conventional CCA and 3 proposed online adaptation methods on the Benchmark Dataset and the UCSD Dataset. Fig. S1(a) and (e) provide the comparisons between the conventional CCA and the online msCCA. Although the online msCCA has the least performance in the proposed methods, it still outperforms the conventional CCA in most cases. Comparing Fig. S1(a) and (e), the improvements of the online msCCA are not very obvious on the UCSD Dataset. Despite this, when the signal lengths can make the average ITRs achieve their maximum values, the

online msCCA can still perform much better than the conventional CCA for most individuals.

Fig. S1(b), (f), (c) and (g) illustrate the comparisons between the online msCCA and the online msTRCA as well as the online msCCA and the online TDCA. The results further confirm that the online TDCA and the online msTRCA generally achieve higher performance than the online msCCA. Fig. S1(d) and (h) directly compare the performance of the online msTRCA and the online TDCA. While their performance is similar in most cases, the online TDCA provides a marked advantage where online msTRCA performs poorly.

S.III. Detailed recognition performance of deep learning-based SSVEP recognition methods without sufficient calibration

Fig. S2 compares the recognition performance between three proposed online adaptation methods and four deep learning-based methods, i.e., EEGNet, Conv-CA, Guney-DNN, CCNN. These deep learning-based methods are pre-calibrated by individual 1 block or 2 blocks calibration data. We can observe that, since one or two individual blocks are seriously insufficient for training these deep learning models, the performance of these deep learning-based methods is very low. Among these deep learning-based methods, when utilizing more calibration data, the performance will be increased. In addition, under such insufficient calibration cases, the CCNN can perform best among these deep learning-based methods. However, even for 2 blocks of calibration data, the CCNN still significantly performs worse than the three proposed online adaptation methods. It should be noticed that, unlike these deep learning-based methods, the proposed online adaptation methods do not require any pre-calibration.

References

[S1] Z. Wang, L. Shen, Y. Yang, Y. Ma, C. M. Wong, Z. Liu, C. Lin, C. T. Hon, T. Qian, and F. Wan, "A least-square unified framework for spatial filtering in SSVEP-based BCIs," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 32, pp. 2470–2481, 2024.

[S2] C. M. Wong, Z. Wang, M. Nakanishi, B. Wang, A. Rosa, P. Chen, T.-P. Jung, and F. Wan, "Online adaptation boosts SSVEP-based BCI performance," *IEEE Trans. Biomed. Eng.*, vol. 69, pp. 2018–2028, 2022.